INCH-POUND

MIL-DTL-25666F (USAF)
15 September 1995
SUPERSEDING
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03 June 1980

DETAIL SPECIFICATION

CONVERTER, LIQUID OXYGEN, CAPACITANCE TYPE GAUGING,

GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for liquid oxygen converters that will store liquid oxygen on board an aircraft and that will convert the liquid oxygen to gas as required to supply the oxygen regulators. It includes converters of various capacities and operating pressures with either integral warming coils or separate aircraft installed warming coils. Provisions for quick-change installation in aircraft or permanent installation in aircraft are included.

1.2 Classification. Classification of the converters are as follows:

Associated specification	Type	Capacity Liters	Operating pressure PSI
MIL-DTL-25666/1	GCU-2A/A	10	70
MIL-DTL-25666/2	GCU-3/A	10	70
MIL-DTL-25666/3	GCU-10/A	10	300
MIL-DTL-25666/4	GCU-12A/A	5	70
MIL-DTL-25666/5	GCU-17/A	25	300
MIL-DTL-25666/6	GCU-20/A	75	300
MIL-DTL-25666/7	ME-3	25	300

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-3037 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660
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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index for Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical

DEPARTMENT OF DEFENSE

MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard Optimum Selected Series: General Requirements for
MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-F-18264	Finishes: Organic, Weapons System, Application and Control of
MIL-D-19326	Design and Installation of Liquid Oxygen Systems in Aircraft, General Specification for
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-L-25567	Leak Detection Compound, Oxygen Systems
MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specification for
MIL-V-25961	Valve, Fill-Buildup-Vent, Liquid Oxygen Converter CRU-50/A
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-T-27730	Tape, Antiseize, Polytetrafluoroethylene, with Dispenser
MIL-V-38201	Valve, Filler, Liquid Oxygen, Female CRU-59/E

STANDARDS

FEDERAL

FED-STD-595	Colors Used in Government Procurement

DEPARTMENT OF DEFENSE

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-721	Definitions of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Testing For Engineering Development, Qualification and Production
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-1359	Cleaning Methods and Procedures for Breathing Oxygen Equipment
MS33583	Tubing End Double Flare, Standard Dimensions for
MS33584	Tubing End Standard Dimensions for Flared

(Unless otherwise indicated, copies of the above specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NHB 8060.1	Office of Safety and Mission Quality Flammability, Odor, Offgassing, and Compatibility Requirements and Test Procedures for Materials in Environments that Support Combustion

(Application for copies should be addressed to the National Aeronautics and Space and Administration (NASA), NASA Headquarters, Technical Standards Division, Code QE, Washington, DC 20546)

2.3 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2512	Compatibility of Materials with Liquid Oxygen (Impact Sensitivity Threshold and Pass-Fail Technique), Test Method for
ASTM B117	Standard Practice for Operating Salt Spray (Fog) Testing Apparatus

(Application for copies should be addressed to the American Society For Testing Materials, 1916 Race St, Philadelphia, PA 19103-1187)

METALS AND ALLOYS IN THE UNIFIED NUMBERING SYSTEM (UNS)

UNS S30403	Austenitic Cr-Ni Stainless Steel (Low Carbon)

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS standards), the text of this documents takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be as specified herein and in accordance with the applicable associated specification. In the event of any conflict between the requirements of this specification and the associated specification, the latter shall govern.

3.1.1 Qualification. The converters furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Major components. The major components of the converters shall be as follows:

a. Liquid oxygen container (see 3.5.1).

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b. Capacity sensing element and connections (see 3.5.2).

c. Fill-buildup-vent valve (applicable to converters with provisions for quick-change installations)(see 3.5.3).

d. Pressure closing valve (see 3.5.4).

e. Pressure relief valve (see 3.5.5).

f. Overpressure safety device (see 3.5.6).

g. Mounting provision and system connections (see 3.5.7).

3.3 Materials. Materials shall conform to applicable specifications as identified herein. Materials that are not covered by specifications shall be suitable for use in liquid oxygen systems. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the function of the end item.

3.3.1 Metals. Metals shall be corrosion-resistant steel or nonferrous metal that has been treated to resist corrosion. Metallic materials that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions likely to occur during the usage of the converter shall be protected against such corrosion in a manner that will not in any way prevent compliance with the performance requirements of this specification.

3.3.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact.

3.3.1.2 Nonmetallic materials. Nonmetallic materials shall have the physical properties required for the converter to function, without being adversely affected, while exposed to cryogenic temperatures. The contractor shall be responsible for the evaluation of materials for their suitability in each application. The contractor may evaluate materials in accordance with either ASTM D2512 or NASA NHB 8060.1.

3.3.2 Exterior coating. After the cleaning specified (see 3.9) has been completed, the assembled liquid oxygen container and any parts of the other components that are to be painted shall be coated with a primer compatible with the finish to be applied. One or more coats of durable, high quality paint shall be applied as required to achieve a smooth protective finish free of cracks, checks, or runs. The color of the topcoat shall conform to 14187 of FED-STD-595. Organic finishes shall be applied in accordance with MIL-F-18264.

3.4 Design and construction. Each converter shall be designed and constructed in accordance with the applicable associated specification (see 1.2).

3.4.1 Tubing and fittings. All openings of the finished converter shall be capped or plugged to prevent thread damage and to prevent the entry of moisture and foreign matter. Aluminum tubing that interconnects components shall be double flared to conform to MS33583. Tube fittings and pipe thread bosses for mating the converter to aircraft lines shall be corrosion-resistant steel. Corrosion-resistant steel tubing shall be single flared to conform to MS33584.

3.4.2 Antiseize tape. Antiseize tape that conforms to MIL-T-27730 shall be used on tapered male pipe thread fittings. The antiseize tape shall not be used on the straight threads or the tubing fittings. Particles of the antiseize tape shall not be allowed to enter the inside of plumbing lines.

3.4.3 Check valves and flow restructures. Check valves and flow restrictors may be used in the circuit as required for function. Check valves and flow restrictors shall be spring loaded as necessary to avoid any possibility of entrapment of liquid oxygen in a dead end circuit.

3.4.4 Dimensions. The dimensions of the converter shall be as specified in the applicable associated specification.

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3.4.5 Threads. Screw threads shall be in accordance with MIL-S-7742. Pipe threads shall be in accordance with MIL-P-7105.

3.4.6 Weight. The weight of the complete converter, including the converter portion of any quick disconnects, shall not exceed the weight specified in the applicable associated specification.

3.5 Details of components.

3.5.1 Liquid oxygen container. The liquid oxygen container shall be a double-walled, vacuum-insulated container that will limit heat transfer and maintain a near perfect permanent vacuum. The vacuum between the inner wall (inner vessel) and the outer wall (outer vessel) shall contain a getter (see 6.5) that will not propagate fire or explosion in the event of a direct hit by an incendiary projectile. The fill line entrance to the storage space of a liquid oxygen container shall not allow the filling stream to impinge on the vent outlet opening. The inner vessel and the outer vessel shall be fabricated utilizing manufacturing processes and selected corrosion-resistant steels that conform to S30403 to assure limited chromium carbide grain formulation thereby reducing susceptibility to stress corrosion. The outer vessel shall have a rupture area that is positioned or shielded to minimize the danger to nearby personnel.

3.5.2 Capacity sensing element and connections. A capacity sensing element that will provide a linear incremental increase in electrical capacitance proportional to the liquid oxygen quantity shall be incorporated within the liquid oxygen container. The capacity sensing element shall function with gage system components that conform to MIL-I-25645. The two leads from the capacity sensing element shall terminate in electrical receptacles. The electrical receptacle for the high impedance lead shall conform to style B of MIL-C-25516. The electrical receptacle for the low impedance lead shall conform to style E of MIL-C-25516. The electrical receptacles of the converters shall be provided with storage caps that preclude contamination of the receptacle terminals during transit and storage.

3.5.3 Fill-buildup-vent valve. When the applicable associated specification specifies a fill-buildup-vent valve as a component of the converter, the fill-buildup-vent valve shall meet the requirements specified in MIL-V-25961. The fill-buildup-vent valve shall be located so that it will interface and provide the hand operation clearance necessary to accomplish normal filling actions with a CRU-59/E female filler valve that conforms to MIL-V-38201.

3.5.4 Pressure closing valve. The pressure closing valve shall be actuated by a leakproof metal bellows as the pressure responsive element. The adjustment feature of the pressure closing valve shall be secured in place by safety wire or sealant, or both, to prevent vibration induced changes.

3.5.5 Pressure relief valve. The pressure relief valve or valves, shall meet the performance requirements of MIL-V-9050. The type of pressure relief valve or valves identified in the applicable associated specification shall be utilized.

3.5.6 Overpressure safety device. A secondary overpressure device shall be incorporated in the plumbing of the converter at a location to insure rapid dumping of the head pressure from the converter in the event the pressure relief valve fails to open.

3.5.7 Mounting provisions and system connections. The mounting provisions for either quick-change or fixed installation type converters and their system connections shall be as specified in the applicable associated specification.

3.6 Interchangeability. All parts having the same manufacturer's part numbers shall be dimensionally and functionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall apply.

3.7 Marking.

3.7.1 Identification of product. Each converter shall be permanently identified as specified in MIL-STD-130.

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3.7.2 Precautionary marking. Each converter shall have a decal or a plate that is located in a conspicuous place. The following information shall be marked on the decal or plate.

WARNING
HIGH VACUUM CONTAINER
HANDLE WITH CARE

3.8 Workmanship. The converters shall be manufactured in accordance with high grade aircraft equipment manufacturing practices. The converters shall be thoroughly cleaned of oil and any other foreign matter and shall be free of defects that might affect appearance or functionality.

3.9 Cleaning.

3.9.1 Preassembly of converter. Prior to the assembly of the converter, the internal and external surfaces of the components shall be degreased by ultrasonics, by immersion, scrubbing, or by pressure spraying with cleaning compounds in accordance with MIL-STD-1359 or qualifying activity approval of the contractor's procedure.

3.9.2 After welding and brazing. After all welding and brazing operations have been completed, all weld areas shall be cleaned, and weld spatter shall be removed by suitable mechanical or abrasive cleaning techniques. All flux residue shall be thoroughly removed in order to preclude corrosive effects which may occur undetected beneath the paint and result in early failure of the converter in service. The cleaning solutions and the processes that are used shall be compatible with the converter and its expected service life.

3.10 Performance.

3.10.1 Final assembly pressure. When the converter is subjected to the final assembly pressure specified in the applicable associated specification, it shall not leak except for the leakage allowed for installed components by their respective specifications. The specified primary pressure relief valve shall vent the converter at the flow rate established in MIL-V-9050 in response to an internal pressure not greater than 10 psig above the maximum normal working pressure of that valve. The primary relief valve is the component relief valve that has the lowest operating pressure range (see 4.6.1).

3.10.2 Filling time. When liquid oxygen is transferred from standard equipment to the converter, the time required to fill the converter shall not exceed the filling time specified in the applicable associated specification (see 4.6.2).

3.10.3 Evaporation loss.

3.10.3.1 Vent condition. The evaporation loss rate of the essentially full converter in the vent condition shall not exceed the maximum evaporation loss specified in the applicable associated specification (see 4.6.3.1).

3.10.3.2 Buildup condition. The evaporation loss rate of the essentially filled converter in the buildup condition shall not exceed the maximum evaporation loss specified in the applicable associated specification (see 4.6.3.2).

3.10.4 Buildup time. The time required for the converter to reach its operating pressure immediately after it has been filled shall not exceed the buildup time specified in the applicable associated specification (see 4.6.4).

3.10.5 Dielectric strength. When tested as specified (see 4.6.5), the current leakage shall not exceed 1 milliampere.

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3.10.6 Insulation resistance. When tested as specified (see 4.6.6) the insulation resistance between the two lead wires of the sensing element in the converter, shall not be less than 20 megohms. When tested as specified (see 4.6.6) the insulation resistance between each lead wire of the sensing element in the converter and the converter ground, shall not be less than 10 megohms.

3.10.7 Capacitance. The capacitance of the sensing element, after installation in the empty converter, shall be within the limits specified in the applicable associated specification (see 4.6.7).

3.10.8 Inner vessel pressure.

3.10.8.1 Inner vessel pressure. When the inner vessel is subjected to the pressure specified in the associated specification, it shall not show any evidence of material failure (see 4.6.8.1).

3.10.8.2 Permanent volumetric expansion. The permanent volumetric expansion of the inner vessel shall not exceed 10% of the total volumetric expansion while the inner vessel is subjected to the pressure specified in the associated specification (see 4.6.8.2).

3.10.9 Pressure closing valve leakage. When the converter buildup pressure conditions are applied to the pressure closing valve, the leakage past the valve seat shall not exceed 0.02 liter per minute measured at normal temperature and pressure, with a pressure differential of 5 to 10 psi being maintained across the pressure closing valve seat (see 4.6.9).

3.10.10 Delivery rate.

3.10.10.1 Constant flow. The converter shall deliver oxygen at a constant flow rate to provide the mass flow and delivery pressure specified in the applicable associated specification (see 4.6.10.1).

3.10.10.2 Cyclic flow. The converter shall deliver oxygen at a cyclic flow rate to provide the mass flow and delivery pressure specified in the applicable associated specification (see 4.6.10.2).

3.10.11 Calibration. The capacitance values representing the remaining quantity of liquid oxygen shall establish an essentially constant slope and shall be within the range specified in the applicable associated specification (see 4.6.11).

3.10.12 Operating temperature. The converter shall provide the required delivery rate (see 3.10.10) at the required ambient temperature (see 4.6.12).

3.10.13 Overpressure safety. The overpressure safety device shall remain leaktight under all normal use conditions. The overpressure safety device shall open at any pressure level between 1.4 and 2.5 times the maximum opening pressure of the primary pressure relief valve but in all cases at a pressure low enough to preclude failure of the liquid oxygen container (see 4.6.13).

3.10.14 Rupture area. The outer shell of the container shall be provided with a rupture point having not less than a 0.375 (9.525 mm) diameter escape area. Pressure required to rupture the escape area shall be at least 20% less than the burst pressure of the outer container shell. The rupture point shall be located in such a position that in case of rupture the danger to personnel will be at a minimum (see 4.6.14).

3.10.15 Vibration.

3.10.15.1 Vibration. The converter, during and after exposure to vibration (see 4.6.15.1)., shall function within the limits established in the applicable associated specification.

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3.10.15.2 Vibration - resonance. The converter shall not exhibit physical evidence of damage after completion of the resonance portion of the vibration test (see 4.6.15.2). The converter shall be physically suitable to complete the remainder of the tests.

3.10.15.3 Vibration - cycling. The converter shall not exhibit physical evidence of damage after completion of the cycling portion of the vibration test (see 4.6.15.3). The converter shall be physically suitable to complete the remainder of the tests.

3.10.16 Acceleration. After the filled converter has been subjected to acceleration (see 4.6.16), the converter shall meet the requirements of final assembly pressure (see 3.10.1), evaporation loss buildup condition (see 3.10.3.2) and delivery rate (see 3.10.10). The delivery rate test shall be limited to 20 minutes after this acceleration test.

3.10.17 Salt fog. Exposure of the converter to salt fog (see 4.6.17) shall not cause corrosion indicative of dissimilar metals or corrosion that could preclude normal operation of the converter or create any unsafe conditions. After the salt deposits are removed from the converter, the converter shall meet the requirements specified (see 3.10.1, 3.10.3.2, 3.10.5, 3.10.6, 3.10.7, 3.10.10.1 and 3.10.11).

3.10.18 Gunfire. When the section of the converter that contains the liquid oxygen is pierced by a gun-fired projectile, the converter shall not shatter (see 6.5) and shall remain in one piece except for small pieces torn from or broken off the converter by the projectile. The mounting provisions of the converter shall withstand the forces of impact of the gun-fired projectile and the discharge of the contents of the converter (see 4.6.18).

3.10.19 Reliability. When the converters are operated in a manner simulating actual use in aircraft, the converters shall demonstrate a mean time between failures (MTBF) of not less than 300 hours. A failure shall be identified as any malfunction that causes the converter to be unable to deliver the rated flow, unable to maintain the specified operating pressure, or unable to demonstrate acceptable loss rates while on standby. The accept-reject criteria for the reliability testing shall be in accordance with test plan I (10% decision risk) of MIL-STD-781 (see 4.6.19).

3.10.19.1 Service life. The service life of the converters, which is the period of use during which the MTBF remains essentially constant, shall be not less than 1,000 operational hours, excluding standby periods when no flow is being taken from the supply connection.

3.10.20 Maintainability. The corrective maintenance as defined in MIL-STD-721 shall not exceed 2 hours mean maintenance time to repair the plumbing leaks; to remove and replace the components such as the pressure closing valve, the relief valve, or any overpressure safety device; or to replace the tubing runs and the mounting hardware. The loss of vacuum or capacitance gage malfunctions, which require the replacement of the entire liquid oxygen container, shall be excluded from this requirement (see 4.6.20).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.4).

4.2. Qualification inspection. Qualification inspection shall include all the inspections, tests, and verification of this specification and the applicable associated specification.

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4.2.1 Qualification samples. The manufacturer shall use two converters for conducting the qualification inspection. One converter shall be subjected to the vibration (see 4.6.15), acceleration (see 4.6.16) and gunfire (see 4.6.18) tests and the second unit shall be subjected to the remainder of the tests specified (see 4.2). This second unit shall have the salt fog test (see 4.6.17) performed next to last and the reliability test (see 4.6.19) performed last.

4.2.2 Qualification sample for the qualifying activity. Along with the qualification test report, the manufacturer shall submit one converter to the qualifying activity. This sample may be a tested sample, except that it shall not be subjected to the gunfire test.

4.3 Inspection conditions.

4.3.1 Environmental conditions. Unless otherwise specified, all tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure at the time each test is accomplished shall be recorded. This information shall be available for computation of corrections, where required, to normal temperature and pressure conditions. Normal temperature and pressure conditions are 14.7 psi absolute and 70° F. Test instruments shall be calibrated or adjusted according to their required usage in conducting the tests specified herein.

4.3.2 Master capacitance bridge. The master capacitance bridge that is used to test the sensing element shall have an accuracy equal to or better than ±0.2 percent of the nominal full converter capacitance or 0.2 picofarad, whichever is larger. The accuracy of the master capacitance bridge used for the measurements specified herein shall be applied to broaden the acceptable capacitance range specified in the applicable associated specification.

4.3.3 Cryogenic liquids and gases. Unless otherwise specified gaseous or liquid oxygen conforming to MIL-O-27210, type I or type II shall be used for all tests. If the use of oxygen for the tests specified herein is hazardous because of the nature of the test area or equipment, liquid or gaseous nitrogen that conforms to BB-N-411 may be used.

4.3.4 Post test cleanup. Upon the conclusion of testing in an unclean area any foreign material contamination shall be removed before resuming testing with oxygen. All traces of leak detection compound shall be removed from the converter after each leakage test.

4.3.5 Purging. The completed converter, prior to preparation for delivery, shall be purged with hot, dry oxygen, conforming to MIL-O-27210, type I or nitrogen conforming to BB-N-411, type I class 1, grade B, until all traces of the contaminant are removed. The temperature of the purging gas should approach but not exceed, 250°F (121°C) at the converter inlet. The purge pressure shall be as necessary to create sufficient flow to cause the converter internal surfaces to rise to a temperature that will expedite contaminant removal, but should not exceed 100 psig (690 kPa). After purging, charge the container with gaseous oxygen at 25 ±5 psig (172 ±34 kPa) and close all openings with approved closures.

4.3.6 Cleanliness verification. Components of the converter shall be inspected for cleaning in accordance with MIL-STD-1359 or qualifying activity approval of the contractor's procedure.

4.4 Conformance inspection. The conformance inspection shall consist of the following:

a. Individual inspection.

b. Sampling inspection.

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4.4.1 Individual inspection.

4.4.1.1 Converters. Each converter shall be subjected to the following methods of inspection:

a. Visual examination (see 4.5.1).

b. Final assembly pressure (see 4.6.1).

c. Filling time (see 4.6.2).

d. Evaporation loss (see 4.6.3).

e. Buildup time (see 4.6.4).

f. Dielectric strength (see 4.6.5).

g. Insulation resistance (see 4.6.6).

h. Capacitance (see 4.6.7).

i. Delivery rate (see 4.6.10), the delivery rate shall be monitored for 20 minutes and then the test shall be terminated.

4.4.1.2 Inner vessels. Before the outer vessel is added to the inner vessel, each assembled inner vessel complete with all lines and fittings shall be subjected to the pressure test described (see 4.6.8.1).

4.4.1.3 Pressure closing valves. Before the pressure closing valves are installed in the buildup circuit of the converters, each pressure closing valve shall be subjected to the pressure closing valve leakage test described (see 4.6.9).

4.4.2 Sampling inspection. The lot size for sampling plan inspection shall be 50 units (converters). One out of every 50 units that have been manufactured under essentially the same conditions using the same facilities and equipment shall be selected for sampling inspection. Continuous production is not required. Component parts and material need not be delivered from the same batch or lot. The following tests or inspections are required:

a. Dimensional examination (see 4.5.2).

b. Operating temperature test (see 4.6.12).

c. Calibration (see 4.6.11).

d. Overpressure safety (4.6.13).

e. Delivery rate (see 4.6.10).

f. Permanent volumetric expansion (see 4.6.8.2).

4.4.2.1 Rejection and retest. If one or more items selected from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative and notify the procuring activity in writing the cause of failure, the action taken to preclude recurrence, and the impact this failure might have on scheduled deliveries. After correction, all of the sampling inspection shall be repeated.

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4.4.2.2 Continuation of inspection. For production reasons, individual inspection or other sampling plans may be continued pending the investigation of a sampling inspection failure. Final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of this specification.

4.4.3 Defects in items already accepted. The investigation of a test failure could indicate that defects might exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.5 Examination of product.

4.5.1 Visual examination. Every converter shall be visually examined to determine conformance to the requirements of this specification and the applicable associated specification. The classification of defects shall be according to table I unless otherwise specified. All defects not corrected shall be cause for rejection. Minor color shade deviation need not be corrected.

TABLE I. Classification of defects for visual examination of the converter.

Critical	Minor
1. Material imperfections - foreign matter embedded 2. Surface - unclean, rough, misaligned or containing cracks, nicks or other flaws. 3. Any component missing, malformed, fractured or otherwise damaged. 4. Any component loose or otherwise not securely retained. 5. Incorrect assembling or improper positioning of components. 6. Any functioning part that works with difficulty. 7. Faulty workmanship or other irregularities. 8. Marking - missing, insufficient, incorrect, illegible or not permanent.	201. Color shade not as specified.

4.5.2 Dimensional examination. The converter shall be examined to determine conformance to the dimensions specified in the applicable associated specification. If any dimension of the converter is not as specified, the converter shall be rejected.

4.6 Performance.

4.6.1 Final assembly pressure. The converter, including all its fittings and connections, shall be subjected to the pneumatic pressure that is specified as the final assembly pressure in the applicable associated specification. While this pressure is maintained, the converter shall be tested by applying leak detection compound that conforms to MIL-L-25567 to all fittings and connections of the converter and then examining all fittings and connections of the converter for leakage. The pressure shall then be increased to a level sufficient to open the pressure relief valve

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and to demonstrate that the required flow capacity can be achieved without excessive pressure. If there is any leakage from the converter, except for the leakages that are allowed for installed components by their respective specifications, the converter shall be rejected (see 3.10.1). Nitrogen, per BB-N-411, type I, grade B, class 1 may be used in lieu of oxygen for the inspection.

4.6.2 Filling time. With the converter at approximately 70° F, the time required to fill the converter by transfer of liquid oxygen from a storage container to the converter shall be determined. The pressure at the converter inlet shall be maintained at 20 psig unless the converter has a fill buildup vent valve, then the pressure shall be 30 psig. The converter shall be considered full when it contains at least the nominal capacity of the liquid oxygen, determined by weight. The time required to fill the converter shall not exceed that specified (see 3.10.2).

4.6.3 Evaporation loss.

4.6.3.1 Vent condition. The evaporation loss rate in the vent condition for a 24 hour period shall be determined two times as specified herein. The first (initial) determination may be made anytime after evacuation and pinch-off of the liquid oxygen container. The second determination shall be made after a lapse of at least 30 days from the date of the first determination. If the first or the second determination exceeds the evaporation loss specified in the applicable associated specification, the liquid oxygen container shall be rejected. The evaporation rate shall be determined as follows: The container shall be filled with liquid oxygen to the rated capacity and allowed to stand in the vented condition for a minimum of 2 hours for equilibrium/stabilization of the liquid oxygen to be established. After which, with the unit vented to the atmosphere, the loss in weight shall be at a rate not to exceed the requirements specified (see 3.10.3.1). The loss in weight for a 24 hour period shall be recorded.

4.6.3.2 Buildup condition. Upon completion of the vented evaporation loss tests, the complete converter assembly including all components shall be subjected to a 24 hour evaporation loss test in the pressurized condition. The converter shall be placed in the pressurized condition and the initial weight reading shall be taken immediately after the converter has been filled. The loss in weight for a 24 hour period shall be recorded. The converter assembly shall pass the requirements specified (see 3.10.3.2).

4.6.4 Buildup time. The converter shall be filled with liquid oxygen and, without delay, shall be set up to generate the operating pressure specified (see 3.10.4). If the time required for the converter to reach its normal operating pressure range exceeds the buildup time specified in the applicable associated specification, the converter shall be rejected.

4.6.5 Dielectric strength. The dielectric strength of the sensing element shall be measured at a potential of 500 volts rms, 60 Hertz. This potential shall be maintained for a period of 10 seconds between each lead wire and the converter ground and between the two lead wires. If the current leakage measured at this potential exceeds that specified (see 3.10.5), the converter shall be rejected.

4.6.6 Insulation resistance. When a potential of 50 volts direct current is applied between the two lead wires of the sensing element in the empty converter, the insulation resistance shall be as specified (see 3.10.6). When a potential of 50 volts direct current is applied between each lead wire of the sensing element in the empty converter and the converter ground, the insulation resistance between either lead wire and the converter ground shall be as specified (see 3.10.6).

4.6.7 Capacitance. With the converter empty, equalized at room temperature, and dry, the capacitance of the sensing element shall be measured with the master capacitance bridge. If the capacitance of the sensing element is not within the range specified (see 3.10.7), the converter shall be rejected.

4.6.8 Inner vessel pressure.

4.6.8.1 Inner vessel pressure. This test shall be conducted before the outer vessel is added. The inner vessel shall be subjected to the inner vessel pressure specified (see 3.10.8.1) for at least 1 minute. The inner vessel shall not show any evidence of material failure.

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4.6.8.2 Permanent volumetric expansion. Any internal pressure that has been applied to the inner vessel before the permanent volumetric expansion test shall not have exceeded 90% of the pressure to be applied during this test. The assembled inner vessel, complete with all lines and fittings, shall be enclosed in a water jacket where the inner vessel will be totally submerged and free to expand in all directions. The inner vessel shall be subjected to the hydrostatic or pneumatic pressure specified (see 3.10.8.2) for at least 1 minute, or longer, to insure the complete expansion of the inner vessel. The difference in temperature of the fluid inside the inner vessel and the fluid outside the inner vessel shall not exceed 5° F. If the test pressure cannot be maintained due to failure of the test apparatus, the permanent volumetric expansion test may be repeated with the test pressure increased by 10%. If the permanent volumetric expansion of the inner vessel exceeds 10% of the total volumetric expansion that existed while the inner vessel retained the specified test pressure, the inner vessel shall be rejected.

4.6.9 Pressure closing valve leakage. A pneumatic pressure source shall be applied to the inlet of the pressure closing valve. Suitable pressure gages and control valves shall be installed in the inlet and outlet of the pressure closing valve. Pressure shall be applied gradually to the inlet of the pressure closing valve until the pressure closing valve closes. The inlet pressure shall be gradually increased until it exceeds the closing pressure by 5 to 10 psig. The allowed leakage shall not exceed the limits specified (see 3.10.9). Nitrogen conforming to BB-N-411, type I, class 1, grade B may be used for this inspection.

4.6.10 Delivery rate.

4.6.10.1 Constant flow. The converter shall be set up to simulate installation in an aircraft. If the converter does not have integral warming coils, the tubing specified in the table entitled "Minimum Length of Tubing Between Converters and First Station" of MIL-D-19326, or an equivalent heat exchanger, shall be added to the supply line. The supply line shall be connected to a metering valve and a suitable flow monitoring instrument, which can be used to observe and set the flow specified in the applicable associated specification. The converter shall be filled, set for pressure buildup, and then operated until empty. The converter shall deliver the flow as specified (see 3.10.10.1).

4.6.10.2 Cyclic flow. For 70 psig nominal systems the converter shall be connected to a simulated breathing apparatus in order to subject the converter to pulsating flows. The breathing apparatus shall provide instantaneous flows with the inhalation period representing 43 ±5% of the time involved. The number of breathing cycles per minute shall be 24 ±2. The breathing flow rates during the inhalation period shall be sufficient to average 72 liters per minute of oxygen measured at 75° ±20°F. The test shall be conducted, for the length of time specified (see 3.10.10.2), with operating pressure remaining between 65 and 90 psig after the first 15 minutes of operation.

4.6.11 Calibration. The converter shall be filled to capacity with liquid oxygen. After the filled converter has been stabilized for a approximately 10 minutes, the capacitance of the sensing element shall be measured by the master capacitance bridge. Five essentially equal increments of liquid oxygen shall be removed from the converter. After the removal of each increment of liquid oxygen from the converter, the capacitance of the sensing element shall be measured with the master capacitance bridge. Compare this value to the amount of liquid oxygen remaining as determined by weighing the converter at each test point. If the capacitance values measured do not remain within the range specified (see 3.10.11), the converter shall be rejected.

4.6.12 Operating temperature.

4.6.12.1 Low temperature. The converter shall be installed in an environmental temperature chamber and connected into a simulated aircraft oxygen system. The converter shall be filled with liquid oxygen or liquid nitrogen and conditioned at -65° ±5° F for at least 3 hours. The delivery rate shall be 80 to 110% of the maximum flow required by the applicable associated specification until the converter is within 5% of empty. If the converter does not deliver the flow of oxygen specified above, after allowing 10 minutes for stabilization, the converter shall be rejected. Momentary pressure drops below the specified normal operating pressure during the first 10 minutes of flow shall be acceptable.

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4.6.12.2 High temperature. The converter shall be installed in an environmental temperature chamber and connected into a simulated aircraft oxygen system. The converter shall be filled with liquid oxygen or liquid nitrogen and conditioned at plus 260° ±5° F for at least 3 hours. The delivery rate shall be 80 to 110% of the maximum flow required by the applicable associated specification until the converter is within 5% of empty. If the converter does not deliver the flow of oxygen specified above, after allowing 10 minutes for stabilization, the converter shall be rejected. Momentary pressure drops below the specified normal operating pressure during the first 10 minutes of flow will be acceptable.

4.6.13 Overpressure safety. The overpressure safety device shall be subjected to gradually increasing pressure. If the overpressure safety device leaks at a pressure below that specified (see 3.10.13) (the pressure range specified in MIL-V-9050 for the primary pressure relief valve) or fails to open at the specified pressure (see 3.10.13), the overpressure safety device shall be rejected.

4.6.14 Rupture area. The assembled outer vessel complete with all lines and fittings shall be pressurized until it fails. If failure does not occur in the rupture area and in a controlled manner to satisfy the requirements specified (see 3.10.14), the components of the outer vessel shall be rejected.

4.6.15 Vibration.

4.6.15.1 Vibration. The converter, filled with liquid oxygen or equivalent, shall be mounted to a rigid test fixture and shall be vibrated as specified (see 4.6.15.2 and 4.6.15.3). The test fixture shall be designed to eliminate or minimize fixture resonances in the test frequency range. Measurements of the all vibratory accelerations or displacements shall be made on the test fixture adjacent to the shock mounts. In addition the evaporation loss of the converter in the buildup condition, determined within 48 hours after completion of vibration testing, shall not exceed the limits specified (see 3.10.3.2).

4.6.15.2 Vibration - resonance. Resonant nodes of the test converter shall be determined by varying the frequency of applied vibration slowly through the specified range at vibratory accelerations (±4%) defined in table I of the associated specification. The vibratory acceleration applied to any converter weighing more than 50 pounds may be decreased by ±1G for each 10 pound increment of weight above 50 pounds to a value no lower than ± 5G. This allowance applies only to the frequency ranges where the G value exceeds 5G. Resonant node searches shall be accomplished along each of the 3 mutually perpendicular axes of vibration. When more than one resonance is encountered along any one axis, vibration testing shall be sustained for a 30 minute time period for each resonant node. If more than 4 resonances are encountered with vibration applied along any 1 axis, the 4 most severe resonances shall be chosen for the test (see 3.10.15.2).

TABLE II Vibration test schedule (per axis).

Number of resonances	0	1	2	3	4
Total vibration time at resonance 1/		30 min	1 hr	1.5 hr	2 hr
Cycling time per axis	3 hr	2.5 hr	2 hr	1.5 hr	1 hr

1/ 30 minutes at each resonance

4.6.15.3 Vibration - cycling. The converter shall be vibrated under the cycling conditions as specified in the applicable associated specification for the cycling time shown in table II. Note that the cycling time for each axis is dependent on the number of resonant nodes found in 4.6.15.2.

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4.6.16 Acceleration. The converter shall be filled, allowed to reach operating pressure under no flow conditions, and then subjected to an acceleration test in accordance with method 513.4, procedure II of MIL-STD-810. In lieu of actual operation with liquid oxygen or liquid nitrogen, the converter may be filled with a fluid to simulate the operating mass and pressurized to the normal operating condition during the acceleration exposure. The forward, backward and lateral forces applied shall be 4G since mounting orientation is not standardized. The vertically downward force applied shall be 9G. The vertically upward force applied shall be 3G (see 3.10.16).

4.6.17 Salt fog. The converter, with its ports adequately plugged, the probe assembled, shall be subjected to the salt fog test specified in ASTM B117. The length of the salt spray test shall be not less than 50 hours. The converter shall pass the requirements specified (see 3.10.17). The converter shall then be subjected to and pass the following tests:

a. Visual examination (see 4.5.1).

b. Final assembly pressure (see 4.6.1).

c. Evaporation loss (see 4.6.3.2).

d. Dielectric strength (see 4.6.5).

e. Insulation resistance (see 4.6.6).

f. Capacitance (see 4.6.7).

g. Delivery rate (see 4.6.10).

h. Calibration (see 4.6.11).

4.6.18 Gunfire. The converter containing all plumbing and components shall be mounted on a frame capable of withstanding the maximum force which the converter will be subjected to by the gunfire test. The converter shall be filled to rated capacity with liquid oxygen and the nominal operating pressure of converter maintained. The converter shall be fired upon from a distance of approximately 50 yards (47.5m) by a .50 caliber armor piercing incendiary bullet which has been tumbled. The bullet shall strike the converter in the liquid phase. There shall be no evidence of shattering, and the converter assembly shall remain in one piece except components in a direct line of fire. Separation of small components from the converter when they are located in the immediate vicinity of the impact area is satisfactory. The converter shall pass the requirements specified (see 3.10.18).

4.6.19 Reliability. The converters shall be operated as specified in the cyclic flow test (see 4.6.10.2. The constant flow test (see 4.6.10.1) shall be substituted when the cyclic flow test is not required in the associated specification. The flow shall be reduced to approximately 20 percent of the rated capacity of the converter. The flow shall be interrupted every 100 hours, and the evaporation loss (buildup condition) test described (see 4.6.3.2) and the constant flow test described (see 4.6.10.1) shall be performed. Purging to remove contaminants shall not be considered to be "failure-repair" action (see 3.10.19).

4.6.20 Maintainability. The manufacturer shall demonstrate that it is feasible to accomplish disassembly and replacement of exterior components in depot level maintenance shops. Specific components mounted by threaded fasteners shall be replaceable with properly fitting parts by a trained technician within the period specified (see 3.10.20).

5. PACKAGING

5.1 Packaging. See applicable associated specification.

5.2 Marking. See applicable associated specification.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The converters covered by this specification are intended for use in storing and pressurizing liquid oxygen in aircraft and subsequently converting the liquid oxygen and delivering it as gaseous oxygen at the rate required during flight. Reference the associated specification for specific aircraft applications.

6..2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-25666 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from OC-ALC/TICLA, 3001 Staff Drive, OK 73145-5990.

6.4 Supersession data.

Superseding documents	Superseded documents
MIL-DTL-25666/1	MIL-C-25777
MIL-DTL-25666/2	MIL-C-25781
MIL-DTL-25666/3	MIL-C-25974
MIL-DTL-25666/4	MIL-C-25973
MIL-DTL-25666/5	MIL-C-27336
MIL-DTL-25666/6	MIL-C-27652
MIL-DTL-25666/7	MIL-C-25674

6.5 Definitions.

Getter A substance introduced into a vacuum chamber to remove traces of gas.

Shatter Fragmentation of basic materials.

6.6 Part or Identifying Number (PIN). The PIN is as follows:

PIN	Type	Capacity Liters	Operating pressure PSI
M25666/1	GCU-2A/A	10	70
M25666/2	GCU-3/A	10	70
M25666/3	GCU-10/A	10	300
M25666/4	GCU-12A/A	5	70
M25666/5	GCU-17/A	25	300
M25666/6	GCU-20/A	75	300
M25666/7	ME-3	25	300

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6.7 Subject term (key word) listing.

Breathing oxygen
Capacitance
Container, inner
Container, outer
Delivery rate
Dewar
Element, sensing
Evaporation loss
Gunfire
LOX
Valve, fill-buildup-vent
Valve, pressure closing
Valve, pressure relief
70 psi
300 psi

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Reviewer:
Air Force - 11

Preparing activity:
Air Force - 71

Agent:
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Project 1660-F673

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-DTL-25666F	950915

3. DOCUMENT TITLE: CONVERTER, LIQUID OXYGEN, CAPACITANCE TYPE GAUGING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

OC-ALC/TICLA ATTN: H. Edwards 3001 Staff Drive, Ste 1AE1/101A Tinker AFB, OK 73145-5990	b. TELEPHONE (Include Area Code) (1) Commercial (405) 736-5960	(2) AUTOVON 336-5960
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